Stockholm University
Department of Meteorology

Modelling of Large-Scale Circulation in Atmosphere and Ocean (MO8004)

Kelvin Waves
Lab 2

May 12, 2026

Author: Fredrik Bergelv (`fredrik.bergelv@live.se`)

Supervisor: Hanna Winge *and* Verena Junge

1 Theory

(i) No equations, what is it? what are the requrmenets?

(ii) Equations. B.C.

iii Write solh

2 Experiments

2.1 Coastal Kelvin Waves

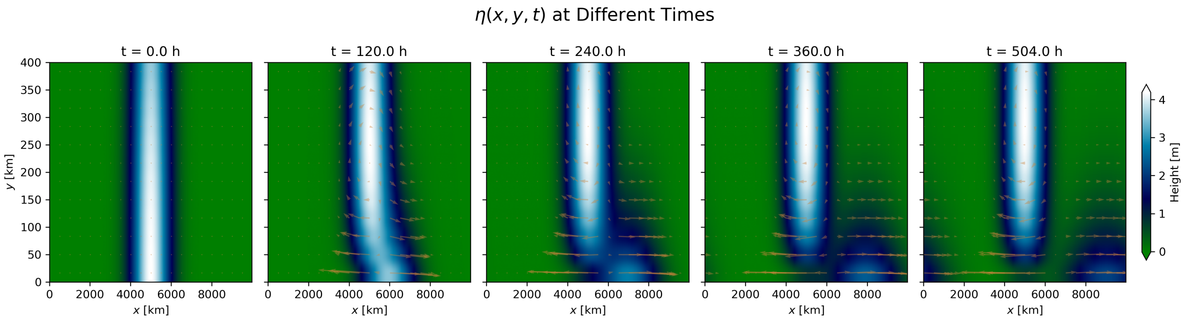

Figure 1: This figure...

Scheme	Time [h]	c_{num} [m/s]	c_{theo} [m/s]	Error [%]
Atlantic	$1 \rightarrow 2$	0.00	9.90	100.00
	$9 \rightarrow 10$	9.26	9.90	6.52
	$23 \rightarrow 24$	9.26	9.90	6.51
	$167 \rightarrow 168$	0.00	9.90	100.00
	$335 \rightarrow 336$	9.26	9.90	6.51
	$503 \rightarrow 504$	0.00	9.90	100.00
	$0 \rightarrow 504$	2.22	9.90	77.56
Atlantic fine grid	$1 \rightarrow 2$	0.00	9.90	100.00
	$9 \rightarrow 10$	4.63	9.90	53.26
	$23 \rightarrow 24$	4.63	9.90	53.26
	$167 \rightarrow 168$	4.63	9.90	53.26
	$335 \rightarrow 336$	4.63	9.90	53.26
	$503 \rightarrow 504$	4.63	9.90	53.26
	$0 \rightarrow 504$	3.56	9.90	64.06
Baltic	$1 \rightarrow 2$	0.00	1.72	100.00
	$9 \rightarrow 10$	0.93	1.72	46.03
	$23 \rightarrow 24$	0.93	1.72	46.03
	$167 \rightarrow 168$	0.93	1.72	46.03
	$335 \rightarrow 336$	0.93	1.72	46.03
	$503 \rightarrow 504$	0.00	1.72	100.00
	$0 \rightarrow 504$	0.55	1.72	67.71
Baltic fine grid	$1 \rightarrow 2$	0.46	1.72	73.01
	$9 \rightarrow 10$	0.93	1.72	46.03
	$23 \rightarrow 24$	0.93	1.72	46.03
	$167 \rightarrow 168$	0.46	1.72	73. 01
	$335 \rightarrow 336$	0.46	1.72	73.01
	$503 \rightarrow 504$	0.93	1.72	46.03
	$0 \rightarrow 504$	0.89	1.72	48.28

Table 1: Instantaneous and average Kelvin wave phase speed compared with theoretical phase speed. The average phase speed can be seen at the bottom of each scheme.

Figure 2: This figure...

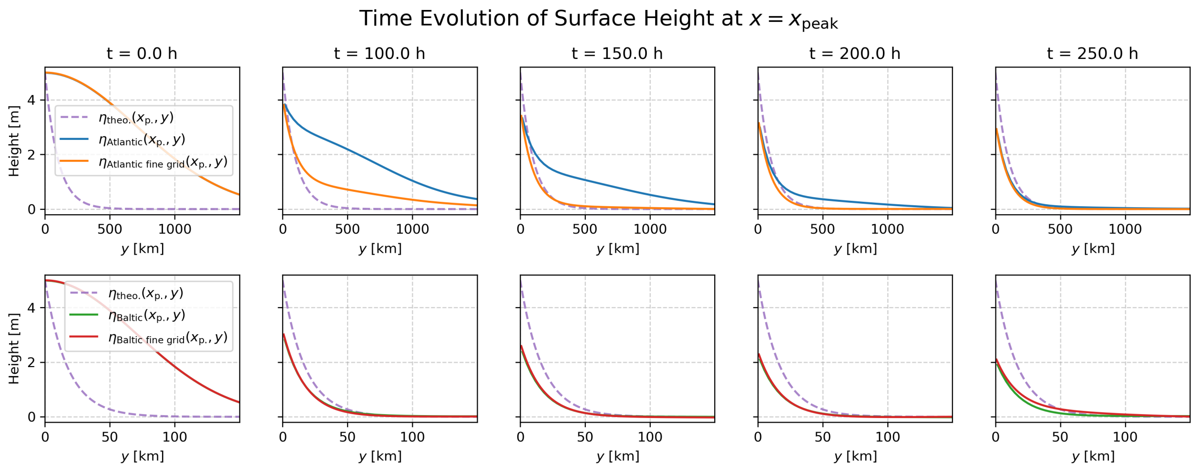

Figure 3: This figure...

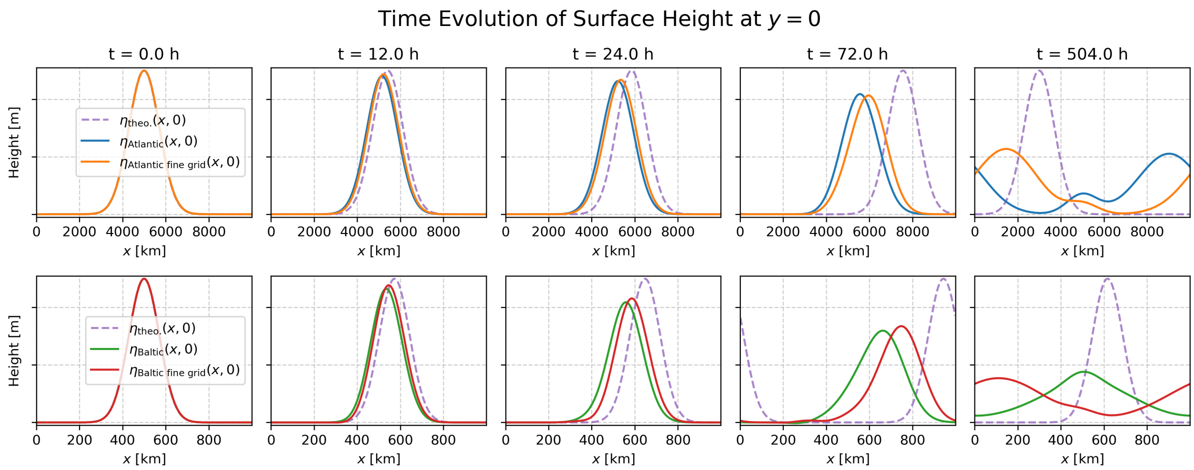

Figure 4: This figure...

2.2 Equatorial Kelvin Waves

(i)

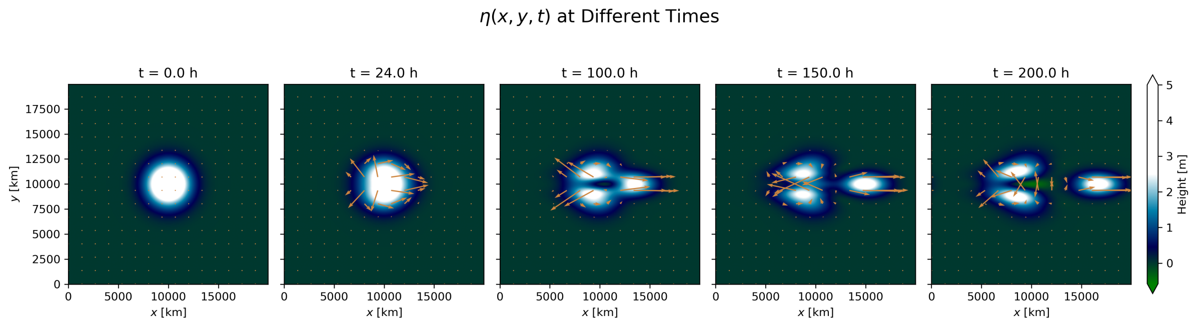

Figure 5: This figure...

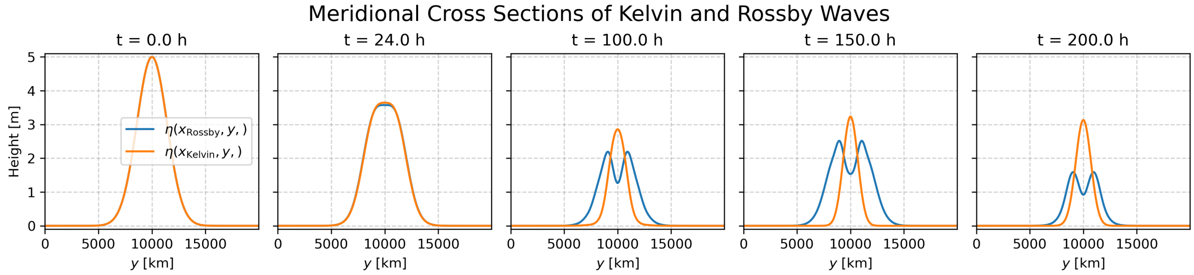

Figure 6: This figure...

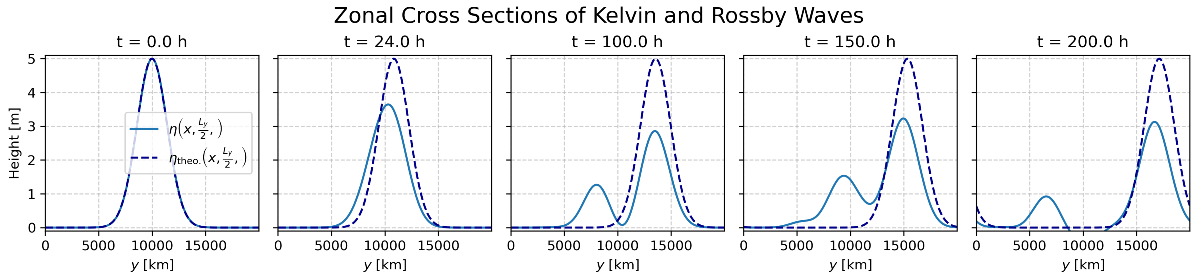

Figure 7: This figure...

Time [h]	x_{max} [km]	c_{num} [m/s]	c_{theo} [m/s]	Error [%]
$0 \to 0$	9966.7	-inf	9.90	nan
$0 \to 24$	10300.0	3.86	9.90	61.05
$24 \to 100$	13500.0	11.70	9.90	18.09
$100 \to 150$	14966.7	8.15	9.90	17.73
$150 \to 200$	16700.0	9.63	9.90	2.78
$0 \to 200$	16700.0	9.33	9.90	5.81

Table 2: Kelvin-wave phase speed diagnostics taken from Figure 7.

A Appendix

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`